Concepts Of Parallel And Distributed Processing

CDA 4125 921175

Fall 2026 - Term

Course Information

Credit Hours: 3.0
Course Description: Concepts Of Parallel And Distributed Processing

This course introduces various systems aspects of parallel and distributed computing. Topics include parallel computer architectures, interconnects, parallel programming paradigms, compilation techniques, runtime libraries, performance evaluation, performance monitoring and tuning, parallel and distributed paradigms and tools for parallel and distributed computing.
Prerequisite: COP3610C

Class Information

Class modality: Face to Face

Class Schedule: Tuesday and Thursday (TR), 2:30-3:45 PM

Location: B103

Faculty Information

Oscar Somarriba

Email: oscar.somarriba@keiseruniversity.edu
Phone: +505-88825648

Credentials

PhD

Office Hours

TR 10 AM-12:30 PM (meeting previously agreed by WhatsApp)-LIBRARY

Program Mission

The Bachelor of Science in Cloud Engineering program at Keiser University is committed to preparing students for dynamic careers in cloud computing by providing a comprehensive, hands-on education in cloud architecture, security, automation, and infrastructure management. Our mission is to equip students with the technical expertise, problem-solving skills, and professional competencies needed to design, deploy, and optimize secure, scalable, and vendor-neutral cloud solutions. Through a rigorous curriculum, industry-relevant training, and a focus on emerging cloud technologies, the program fosters innovation, ethical responsibility, and continuous professional growth. Graduates will be prepared to contribute effectively to the evolving technology landscape, supporting businesses and organizations in their digital transformation efforts while upholding best practices in security, compliance, and cloud governance.

Program Goal(s)

1. Develop Cloud Computing Expertise Equip students with a deep understanding of cloud computing architectures, platforms, and services, enabling them to design, deploy, and optimize vendor-neutral cloud solutions for various business needs.

2. Enhance Technical and Security Proficiency Provide students with hands-on experience in managing cloud networks, implementing security measures, and ensuring compliance with industry standards and best practices in cloud security and governance.

3. Strengthen Infrastructure and Automation Skills Train students to build, manage, and optimize cloud-based infrastructure through automation, virtualization, and enterprise server administration to support mission-critical applications

4. Cultivate Problem-Solving and Analytical Abilities Prepare students to analyze, troubleshoot, and resolve real-world cloud computing challenges, ensuring high availability, scalability, and performance optimization of cloud environments

5. Foster Professional Growth and Communication Skills Enable students to effectively communicate technical concepts, collaborate in team environments, and engage in continuous professional development to advance in cloud engineering and related fields

Program - Student Learning Objectives/Outcomes

- **Objective/Outcome: 1. Apply Cloud Computing Principles and Technologies Students will demonstrate the ability to design, implement, and manage cloud-based solutions across vendor-neutral platforms, ensuring scalability, availability, and efficiency.**
- **Objective/Outcome: 1. Apply Cloud Computing Principles and Technologies Students will demonstrate the ability to design, implement, and manage cloud-based solutions across vendor-neutral platforms, ensuring scalability, availability, and efficiency.**
- **Objective/Outcome: 2. Implement Security, Compliance, and Risk Management in Cloud Environments Students will apply security best practices, compliance frameworks, and risk management strategies to protect cloud-based infrastructures from vulnerabilities and cyber threats.**
- **Objective/Outcome: 2. Implement Security, Compliance, and Risk Management in Cloud Environments Students will apply security best practices, compliance frameworks, and risk management strategies to protect cloud-based infrastructures from vulnerabilities and cyber threats.**
- **Objective/Outcome: 3. Design and Optimize Cloud Infrastructure Using Automation and Virtualization Students will utilize cloud automation tools, virtualization technologies, and optimization techniques to enhance resource utilization and operational efficiency.**
- **Objective/Outcome: 3. Design and Optimize Cloud Infrastructure Using Automation and Virtualization Students will utilize cloud automation tools, virtualization technologies, and optimization techniques to enhance resource utilization and operational efficiency.**
- **Objective/Outcome: 4. Analyze and Troubleshoot Cloud Performance Issues Students will develop problem-solving skills to diagnose, monitor, and resolve cloud infrastructure and application performance issues using industry tools and methodologies.**
- **Objective/Outcome: 4. Analyze and Troubleshoot Cloud Performance Issues Students will develop problem-solving skills to diagnose, monitor, and resolve cloud infrastructure and application**

performance issues using industry tools and methodologies.

- **Objective/Outcome: 5. Communicate Technical Solutions Effectively in a Professional Setting Students will demonstrate the ability to communicate cloud related concepts, solutions, and recommendations effectively to technical and non-technical stakeholders in written and verbal formats.**
- **Objective/Outcome: 5. Communicate Technical Solutions Effectively in a Professional Setting Students will demonstrate the ability to communicate cloud related concepts, solutions, and recommendations effectively to technical and non-technical stakeholders in written and verbal formats.**

Course Learning Objectives

- **Upon completion of the course, students are able to:**
- **1. Describe the storage and use of very large datasets**
- **2. Articulate challenges to operating on large, distributed data**
- **3. Explain parallel programming platforms**
- **4. Explain how distributed systems communicate and share work**
- **5. Design simple parallel algorithms**

Grading and Evaluation Methods

Evaluation	Percent of Total Grade	Due Date
Group Class Project	**15%**	**December 04, 2026**
Mid-term examination	**15%**	**October 15, 2026**
Assignments (cases, problems, other written assignments), Programming assignments, Participation	**40%**	December **04, 2026**
Quizzes	**10%**	December **01, 2026**
Final Exam	**15%**	December **9-16, 2026**
Post Test	**5%**	December **04, 2026**
Total Percentage:	**100%**	

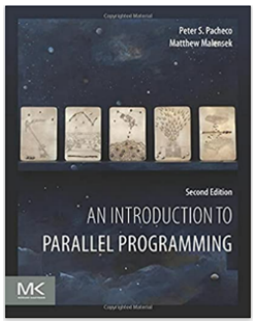
Grading Scale

Letter Grade	Numeric Grade
A	90.00-100.00%
B	80.00-89.99%
C	70.00-79.99%
D	65.00-69.99%
F	Up to 64.99%

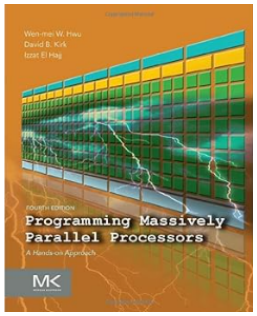
Required Textbook

No Textbook Required

Recommended Resources



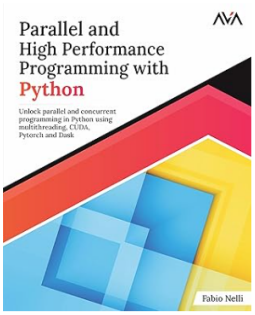
An Introduction to Parallel Programming
ISBN: 978-0128046050
Authors: Petr Pachecho, Matthew Malensek
Publisher: Morgan Kaufmann
Publication Date: 2021
Edition: 2nd. edition



Programming Massively Parallel Processors: A Hands-on Approach
ISBN: 0323912311
Authors: Wen-mei W. Hwu, David B. Kirk, Izzat El Hajj
Publisher: Morgan Kaufmann
Publication Date: 2022
Edition: 4th edition



Introduction to Parallel Computing
ISBN: 9780201648652
Authors: Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar
Publisher: Addison-Wesley
Publication Date: 2003
Edition: 2nd. edition



Parallel and High Performance Programming with Python
Subtitle: Unlock parallel and concurrent programming in Python using multithreading, CUDA, Pytorch and Dask
ISBN: 9388590732
Authors: Fabio Nelli
Publisher: AVA
Publication Date: 2023
Edition: 1st. edition

Topical Outline/Course Assignments/Calendar

Master Syllabus Summary

WEEK 1

Learning Objective(s): 1, 2

Assigned Reading: · Recommneded Textbook (**An Introduction to Parallel Programming**) - Chapters 1, 2

Suggested Learning

Activities: · Hands-on Projects · Intro to Term Projects

Suggested Homework: · Chapter 1 Exercises

Evaluation: · Pre-test, Homework activities and Week 1 test

WEEK 2

Learning Objective(s): 2 and 3

Assigned Reading: · Textbook (**An Introduction to Parallel Programming**) – Chapters 2, 3

Suggested Learning

Activities: · Hands-on Projects · Intro to Term Projects

Suggested Homework: · Chapter 2 Exercises

Evaluation: · Homework activities and mid-term examination

WEEK 3

Week 3, day 1 (September 15, 2026) Central American Independent Holiday

Week 3, day 2, see below:

Learning Objective(s): · 3 and 4

Assigned Reading: · Textbook (**An Introduction to Parallel Programming**) – Chapters 3 and 4

Suggested Learning

Activities: · Hands-on Projects · Term Projects

Suggested Homework: · Chapter 3 Exercises

Evaluation: · Homework activities and Week 3 test

WEEK 4

Week 4, day 1, 8:00am to 9:40am (September 22, 2026) Independent Fest

Week 4, day 2, see below:

Learning Objective(s): · 5

Assigned Reading: · Textbook (**An Introduction to Parallel Programming**) – Chapters 5, 6

Suggested Learning

Activities: · Hands-on Projects · Term Project

Suggested Homework: · Chapter 5 Exercises

Evaluation: · Homework activities, final examination and post-test

Topical Outline/Course Assignments/Calendar in Details (Syllabus)

Week 1, day 1, 2:30pm to 3:45pm introductions, review course syllabus, review class policies, procedures and expectations, take pre-test

Week 1, day 2, 2:30pm to 3:45pm Lecture on chapter1 (book by Pacheco et al.): Why Parallel Computing, classwork1 (Exercises Chapter 1 book by Grama et al.), **Hw1** (Complete Chapter 1 Exercises 1.1-1.5, book by Pacheco et al.), Review question 1 (RQ1)

Week 2, day 1, 2:30pm to 3:45pm Lecture on Chapter 2:Chapter 2 Parallel hardware and Parallel by Pacheco et al., classwork2 (Problem 2.1-2.6, Chapter 2 book by Grama)

Week 2, day 2, 2:30pm to 3:45pm **Hw2** (Complete Chapter 2 Exercises 2.1-2.5, book by Pacheco etal.), RQ2

Week 3, day 1 (September 15, 2026) Central American Independent Holiday.

Week 3, day 2, 2:30pm to 3:45pm Lecture on Chapter 2 final part: Revisiting Parallel program design, Writing and running parallel programs, **Hw3** (Chapter 2 Exercises 2.6-2.15), 2:30-3:45 pm **Programming assignment 1** (**PA1**): Foster's Methodology , RQ3

Week 4, day 1 (September 22, 2026) Independent Fest (Holiday)

Week 4, day 2, 2:30pm to 3:45pm Lecture on Chapter 3 (book by Pacheco et al.): Distributed Memory Programming with MPI) Part 1, Hands-on activity How to do distributed processing with MPI, **Hw4** (Exercises: 3.1 -3.5), RQ4

Week 5, day 1, 2:30pm to 3:45pm Lecture on chapter 3: (book by Pacheco et al.): Distributed Memory Programming with MPI) Part 2, Hands-on activity How to do distributed processing with MPI, **Hw5** (Exercices 3.6-3.10)

Week 5, day 2, 2:30pm to 3:45pm **PA2** (Programming with MPI: 3.1-3.9), RQ5, quiz1 (chapters 1-3)

Week 6, day 1, 2:30pm to 3:45pm Lecture on chapter 4 (book by Pacheco et al.): Shared-memory programming with Pthreads Part 1, Hands-on activity How to do parallel processing with Pthreads

Week 6, day 2, 4:00pm to 5:15pm **Hw6** (Exercises ch4: 4.1-4.10), RQ6

Week 7, day 1,2:30pm to 3:45pm Lecture on chapter 4 (book by Pacheco et al.): Shared-memory programming with Pthreads Part 2, Hands-on activity How to do parallel processing with Pthreads

Week 7, day 2, 2:30pm to 3:45pm **PA3** (**Shared-memory programming with Pthreads**), RQ7, midterm examination

Week 8, day 1, 2:30pm to 3:45pm Lecture on chapter 5 (book by Pacheco et al.): Shared-memory programming with OpenMP, Hands-on activity How to do parallel processing with OpenMP, **Hw7** (Exercises chapter 5 (5.1-5.5))

Week 8, day 2, 2:30pm to 3:45pm **Programming assignment 4 (PA4): Programming with OpenMP**, RQ8

Week 9, day 1, 2:30pm to 3:45pm Lecture on chapter 6 (book by Pacheco et al.) : GPU programming with CUDA, Hands-on activity How to do parallel processing with CUDA 1, **Hw8** (Exercises of chapter 6: 6.1 to 6.10)

Week 9, day 2, 2:30pm to 3:45pm **Programming Assignment 5 (PA5):** 6.1-6.4, Guidelines of the class or course project, RQ9

Week 10, day 1, 2:30pm to 3:45pm Lecture on chapter 7 (book by Pacheco et al):

Parallel Program Development, **Hw9** (Exercises Chapter 7: 7.1-7.5)

Week10, day 2, 2:30pm to 3:45pm **Programming Assignment 6 (PA6):** 7.1-7.6, project registration, RQ10, quiz2 (chapters 4-6)

Week 11, day 1, 2:30pm to 3:45pm Lecture on chapters9 & 16 (book by Wen-mei W. Hwu, et al.): Parallel histogram & Deep Learning, Hands-on activity How to do parallel processing with CUDA 2

Week11, day 2, 2:30pm to 3:45pm **PA7** (Advanced CUDA programming), project progress report, RQ11

Week 12, day 1, 2:30pm to 3:45pm Lecture on chapter 10 (Book by Nelli): Hands-Applications on Parallel Computing with Python, Hands-on activity How to do parallel processing with Parallel Python

Week12, day 2, 2:30pm to 3:45pm **Programming Assignment 8 (PA8): Parallel and Concurrent Programming in Python using Multithreading, CUDA, TensorFlow, and Pytorch,** class project discussion, RQ12

Week 13, day 1, 2:30pm to 3:45pm **Signature Assignment**

Week 13, day 2, Thanksgiving Holiday

Week 14, day 1, 2:30pm to 3:45pm Final project reports

Week 14, day 2, 2:30pm to 3:45pm Posttest, Final Presentations of class project Part 1

Week15, day 1, 2:30pm to 3:45pm Final Presentations of class project Part 2

Week 15, day 2, 2:30pm to 3:45pm Final Presentations of class project Part 3

Week 16, Final exam

Course Guidelines and Policies

Academic Integrity

Students are expected to maintain the highest standards of academic conduct, professional honesty, and personal integrity. Plagiarism, cheating, and other misconduct are serious violations and will not be tolerated, and may result in academic penalties, including suspension or dismissal.

Participation

Participation is a basic requirement for an effective learning community. Students' participation will be assessed and reflected in academic advising sessions and/or student academic progress reports. Class/course participation is based on the following criteria: arriving to class on-time (Face to Face); participation in online chats and engagement in relevant course discussions (Face to Face and On-line); attention during lectures/ discussions/debates; respectful listening when someone else is speaking or providing written responses to content topics/assignment. Respect for your leaning environment, without distractions such as texting, side-bar conversations, checking phones or emails during classroom scheduled times and/or established synchronous learning times is an essential to achieving academic success.

Missed Tests/Quizzes

Makeup exams will be allowed only with pre-approval of the instructor or with an acceptable, documented reason. Acceptable reasons for makeup exams include severe illness, family emergency or other unavoidable events. Exam format for makeup exams may be different than the original exam but the content for the exam will not change.

Late Assignments

Assignments are due at the start of class on or prior to the day noted (or as designated by the instructor). Late assignments without penalty will be accepted only in cases of emergency. Students should discuss turning in late work directly with the instructor and in advance of the due date whenever possible. Late assignments will not be accepted, if the assignment has already been graded and returned to the student. Please note that all work must be submitted by the last day of the class.

Civility/Professionalism

This class is a community of learners, which means we will depend upon each other for support and information. In order to learn, we must be open to the views of people different than ourselves. Please honor the uniqueness of your classmates and appreciate the opportunity we have to learn from one another. Please respect each other's opinions and refrain from personal attacks or demeaning comments of any kind.

It is of the utmost importance to communicate with courtesy and professionalism. Professional courtesy includes respecting other's opinions, being courteous and respectful, and working together in the spirit of cooperation.

Specific Uses of Artificial Intelligence (AI)

There are numerous types of Artificial Intelligence in use today, and it is expected that more will be forthcoming with regularity. For purposes of the University's position, AI is defined as the advancement in technologies that involves many sub-categories of learning or machine-based intelligence, and those that may yet emerge. The position of the university on AI is therefore evolving and subject to change. The university recognizes the range of possibilities for AI to be used positively to enhance learning, as well as opportunities for students to use AI negatively to produce actual work, assignments, projects, and examination responses, in ways not intended by the course instructions or by the professor. The university also recognizes that there are instances in which AI can be used as a teaching tool by professors, for pre-writing, editing, for knowledge, comprehension, and application-based learning, as well as for brainstorming, developing ideas, and many other possibilities. It is critical for the

university to educate and train students in the ability to think critically to perform in their fields of study, requiring originality, competence, creativity, and independence of thought. It is the university's position that all student work should be original, be developed by the student through his or her own critical thought, and that the integrity of academic work in the advancement of knowledge be maintained through such originality in all student work, including online work, in class work, laboratory work, assignments, discussions, projects, and exams. As such, submissions of any work written by, or developed using AI, and/or outside the boundaries of the assignment as described in the syllabus or other formal course instructions, without the professor's explicit knowledge and approval, will be considered a violation of the academic honesty policy. The student must disclose any and all uses of AI in their work, and it is the student's responsibility to provide proof of originality and clear use of appropriate sources, including when requested by the professor for further clarification after submission of work.

University and Program Policies

Students are expected to abide by the policies set forth in the University Undergraduate and Graduate Catalogs and the Student Program Handbook/Manuals. The University Catalogs are available electronically at http://www.keiseruniversity.edu/catalog/. The Program Student Handbook/ Manual is available electronically at the direction of your instructor.

Disability Accommodations

In compliance with the Americans with Disabilities Act (ADA), students who require reasonable accommodations due to a disability to properly execute coursework must complete the application process and receive approval from the review committee. The first step is to consult with the Campus President or Dean of Academic Affairs.